

Creation Date 02-Feb-2010

Revision Date 25-Jan-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	Benzyl chloride
Cat No. :	A12481
Synonyms	alfa-Chlorotoluene
Index No	602-037-00-3
CAS No	100-44-7
EC No	202-853-6
Molecular Formula	C7 H7 Cl
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

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CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Substances/mixtures corrosive to metal

Category 1 (H290)

Health hazards

Acute oral toxicity

Category 4 (H302)

Acute Inhalation Toxicity - Vapors

Category 3 (H331)

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Skin Sensitization

Category 1 (H317)

Germ Cell Mutagenicity

Category 1B (H340)

Carcinogenicity

Category 1B (H350)

Specific target organ toxicity - (single exposure)

Category 3 (H335)

Specific target organ toxicity - (repeated exposure)

Category 2 (H373)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals

H302 - Harmful if swallowed

H331 - Toxic if inhaled

H315 - Causes skin irritation

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H335 - May cause respiratory irritation

H340 - May cause genetic defects

H350 - May cause cancer

H373 - May cause damage to organs through prolonged or repeated exposure

Combustible liquid

Precautionary Statements

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P280 - Wear protective gloves/protective clothing/eye protection/face protection

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2.3. Other hazards

Lachrymator (substance which increases the flow of tears)
This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Benzyl chloride	100-44-7	EEC No. 202-853-6	>95	Met. Corr. 1 (H290) Acute Tox. 4 (H302) Acute Tox. 3 (H331) Skin Irrit. 2 (H315) Eye Dam. 1 (H318) STOT SE 3 (H335) Carc. 1B (H350) STOT RE 2 (H373) Skin Sens. 1 (H317)
Propylene oxide	75-56-9	EEC No. 200-879-2	0.25	Flam. Liq. 1 (H224) Acute Tox. 4 (H302) Acute Tox. 3 (H311) Acute Tox. 3 (H331) Eye Irrit. 2 (H319) STOT SE 3 (H335) Muta. 1B (H340) Carc. 1B (H350)

REACH registration number

-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.
Self-Protection of the First Aider	Use personal protective equipment as required.

4.2. Most important symptoms and effects, both acute and delayed

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. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:
Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling
of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Risk of ignition.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

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7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 6.1C

Switzerland - Storage of hazardous substances

Storage class - SC 6.1
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Benzyl chloride		STEL: 1.5 ppm 15 min STEL: 7.9 mg/m ³ 15 min TWA: 0.5 ppm 8 hr TWA: 2.6 mg/m ³ 8 hr Carc.	TWA / VME: 1 ppm (8 heures). TWA / VME: 5 mg/m ³ (8 heures). STEL / VLCT: 2 ppm. STEL / VLCT: 11 mg/m ³ .	TWA: 1 ppm 8 uren TWA: 5.3 mg/m ³ 8 uren	TWA / VLA-ED: 1 ppm (8 horas) TWA / VLA-ED: 5.3 mg/m ³ (8 horas)
Propylene oxide	TWA: 2.4 mg/m ³ (8h) TWA: 1 ppm (8h)	STEL: 3 ppm 15 min STEL: 7.2 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 2.4 mg/m ³ 8 hr Carc.	TWA / VME: 1 ppm (8 heures). TWA / VME: 2.4 mg/m ³ (8 heures).	TWA: 1 ppm 8 uren TWA: 2.4 mg/m ³ 8 uren	TWA / VLA-ED: 1 ppm (8 horas) TWA / VLA-ED: 2.4 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Benzyl chloride		Haut	TWA: 1 ppm 8 horas		TWA: 0.5 ppm 8 tunteina TWA: 2.6 mg/m ³ 8 tunteina Ceiling: 1.5 ppm Ceiling: 7.9 mg/m ³
Propylene oxide	TWA: 2.4 mg/m ³ 8 ore. Time Weighted Average TWA: 1 ppm 8 ore. Time Weighted Average	TWA: 1 ppm (8 Stunden). AGW - exposure factor 4 TWA: 2.4 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 2 ppm (8 Stunden). MAK TWA: 4.8 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 ppm Höhepunkt: 9.6 mg/m ³	TWA: 1 ppm 8 horas TWA: 2.4 mg/m ³ 8 horas	TWA: 2.4 mg/m ³ 8 uren	TWA: 1 ppm 8 tunteina TWA: 2.4 mg/m ³ 8 tunteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Benzyl chloride	TRK-KZGW: 0.8 mg/m ³ 15 Minuten Haut	Ceiling: 1 ppm Ceiling: 5 mg/m ³	Haut/Peau TWA: 0.2 mg/m ³ 8 Stunden	STEL: 9 mg/m ³ 15 minutach TWA: 3 mg/m ³ 8	Ceiling: 1 ppm Ceiling: 5 mg/m ³

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	TRK-TMW: 0.2 mg/m ³			godzinach	
Propylene oxide	MAK-KZGW: 4 ppm 15 Minuten MAK-KZGW: 8 mg/m ³ 15 Minuten MAK-TMW: 1 ppm 8 Stunden MAK-TMW: 2.4 mg/m ³ 8 Stunden	TWA: 1 ppm 8 timer TWA: 2.4 mg/m ³ 8 timer STEL: 2 ppm 15 minutter STEL: 4.8 mg/m ³ 15 minutter Hud	TWA: 2.5 ppm 8 Stunden TWA: 6 mg/m ³ 8 Stunden	TWA: 2.4 mg/m ³ 8 godzinach	TWA: 1 ppm 8 timer TWA: 2 mg/m ³ 8 timer STEL: 3 ppm 15 minutter. value calculated STEL: 4 mg/m ³ 15 minutter. value calculated Hud

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Benzyl chloride	TWA: 3.0 mg/m ³ STEL : 5.0 mg/m ³	kože TWA-GVI: 0.5 ppm 8 satima. TWA-GVI: 2.6 mg/m ³ 8 satima. STEL-KGVI: 1.5 ppm 15 minutama. STEL-KGVI: 7.9 mg/m ³ 15 minutama.	TWA: 1 ppm 8 hr. STEL: 0.5 ppm 15 min		TWA: 5 mg/m ³ 8 hodinách. Ceiling: 10 mg/m ³
Propylene oxide	TWA: 2.4 mg/m ³ TWA: 1 ppm	TWA-GVI: 1 ppm 8 satima. TWA-GVI: 2.4 mg/m ³ 8 satima.	TWA: 1 ppm 8 hr. TWA: 2.4 mg/m ³ 8 hr. STEL: 3 ppm 15 min STEL: 7.2 mg/m ³ 15 min	TWA: 1 ppm TWA: 2.4 mg/m ³	TWA: 2.4 mg/m ³ 8 hodinách. Ceiling: 5 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Benzyl chloride	TWA: 1 ppm 8 tundides. TWA: 5 mg/m ³ 8 tundides. STEL: 2 ppm 15 minutites. STEL: 11 mg/m ³ 15 minutites.		TWA: 1 ppm TWA: 5 mg/m ³	STEL: 5 mg/m ³ 15 percekben. CK TWA: 2.6 mg/m ³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 1 ppm STEL: 5 mg/m ³
Propylene oxide	TWA: 1 ppm 8 tundides. TWA: 2.4 mg/m ³ 8 tundides. STEL: 10 ppm 15 minutites. STEL: 25 mg/m ³ 15 minutites.		TWA: 1 ppm TWA: 2.4 mg/m ³	TWA: 2.4 mg/m ³ 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 1 ppm 8 klukkustundum. TWA: 2.4 mg/m ³ 8 klukkustundum. Ceiling: 2 ppm Ceiling: 4.8 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Benzyl chloride	TWA: 5 mg/m ³	TWA: 1 ppm IPRD TWA: 5 mg/m ³ IPRD STEL: 2 ppm STEL: 11 mg/m ³			TWA: 1 ppm 8 ore TWA: 5 mg/m ³ 8 ore STEL: 1.5 ppm 15 minute STEL: 8 mg/m ³ 15 minute
Propylene oxide	TWA: 1 ppm TWA: 2.4 mg/m ³	TWA: 1 ppm IPRD TWA: 2.4 mg/m ³ IPRD			TWA: 21 ppm 8 ore TWA: 50 mg/m ³ 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Benzyl chloride	MAC: 0.5 mg/m ³		TWA: 0.2 mg/m ³ 8 urah STEL: 0.8 mg/m ³ 15 minutah	Indicative STEL: 2 ppm 15 minuter Indicative STEL: 11 mg/m ³ 15 minuter TLV: 1 ppm 8 timmar. NGV TLV: 5 mg/m ³ 8 timmar. NGV	
Propylene oxide	Skin notation MAC: 1 mg/m ³	TWA: 2.5 ppm 8 hodinách TWA: 6 mg/m ³ 8 hodinách Potential for cutaneous absorption STEL: 12.5 ppm 15 minútach STEL: 30 mg/m ³ 15 minútach	TWA: 1 ppm 8 urah TWA: 2.4 mg/m ³ 8 urah	Binding STEL: 5 ppm 15 minuter Binding STEL: 12,5 mg/m ³ 15 minuter TLV: 1 ppm 8 timmar. NGV TLV: 2.4 mg/m ³ 8 timmar. NGV	

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Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Component	European Union	United Kingdom	France	Spain	Germany
Propylene oxide					N-(2-Hydroxypropyl)valine: 2500 pmol/g Globin erythrocytes (after at least 3 months exposure)

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Propylene oxide 75-56-9 (0.25)	DNEL = 170mg/m ³		DNEL = 2.4mg/m ³	

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Propylene oxide 75-56-9 (0.25)	PNEC = 0.052mg/L	PNEC = 0.245mg/kg sediment dw	PNEC = 0.52mg/L	PNEC = 10mg/L	PNEC = 0.0186mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Propylene oxide 75-56-9 (0.25)	PNEC = 0.0052mg/L	PNEC = 0.0245mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber Neoprene	See manufacturers recommendations		EN 374	(minimum requirement)

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Natural rubber PVC	-
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Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless - Amber	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	-39 °C / -38.2 °F	
Softening Point	No data available	
Boiling Point/Range	179 °C / 354.2 °F	
Flammability (liquid)	Combustible liquid	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 1.1 Vol% Upper 14 Vol%	
Flash Point	67 °C / 152.6 °F	Method - No information available
Autoignition Temperature	585 - °C / 1085 - °F	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	1.380 mPa.s @ 20°C	
Water Solubility	Moderately soluble	practically insoluble
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Benzyl chloride	2.3	
Propylene oxide	1	
Vapor Pressure	1.2 mbar @ 20°C	
Density / Specific Gravity	1.100	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula C7 H7 Cl

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Molecular Weight 126.59
Explosive Properties explosive air/vapour mixtures possible

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Strong oxidizing agents. Bases. Metals.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂). Hydrogen chloride gas.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral No data available
Dermal No data available
Inhalation No data available

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Benzyl chloride	LD50 = 625 mg/kg (Rat)	-	LC50 = 0.74 mg/L (Rat) 2 h
Propylene oxide	LD50 = 520 mg/kg (Rat)	LD50 = 1244 mg/kg (Rabbit)	9.48 mg/L (Rat) 4 h

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available
May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

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Component	EU	UK	Germany	IARC
Benzyl chloride	Carc Cat. 1B		Cat. 2	Group 2A
Propylene oxide	Carc Cat. 1B			Group 2B

(g) reproductive toxicity;
Reproductive Effects
Developmental Effects
Teratogenicity

No data available
 Experiments have shown reproductive toxicity effects on laboratory animals.
 Developmental effects have occurred in experimental animals.
 Teratogenic effects have occurred in experimental animals.

(h) STOT-single exposure;
Results / Target organs

No data available
 Respiratory system.

(i) STOT-repeated exposure;
Route of exposure
Target Organs

No data available
 Oral
 Heart, Gastrointestinal tract (GI).

(j) aspiration hazard;
 No data available

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.
 Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Do not empty into drains.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Benzyl chloride	LC50: = 4 mg/L, 96h static (Brachydanio rerio) LC50: 4.4 - 5.6 mg/L, 96h static (Pimephales promelas)		
Propylene oxide	LC50: = 215 mg/L, 96h static (Lepomis macrochirus)	EC50: = 350 mg/L, 48h (Daphnia magna)	EC50: = 240 mg/L, 96h (Pseudokirchneriella subcapitata)

Component	Microtox	M-Factor
Benzyl chloride	EC50 = 1.92 mg/L 5 min EC50 = 2.25 mg/L 15 min EC50 = 2.97 mg/L 30 min	
Propylene oxide	EC50 = 3300 mg/L 160 min	

12.2. Persistence and degradability
Persistence

Readily biodegradable
 May persist, based on information available.

12.3. Bioaccumulative potential

May have some potential to bioaccumulate

Component	log Pow	Bioconcentration factor (BCF)
Benzyl chloride	2.3	No data available
Propylene oxide	1	No data available

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12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and sinks in water The product evaporates slowly . Is not likely mobile in the environment due its low water solubility.
Spillage unlikely to penetrate soil

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number

UN1738

14.2. UN proper shipping name

BENZYL CHLORIDE

14.3. Transport hazard class(es)

6.1

Subsidiary Hazard Class

8

14.4. Packing group

II

ADR

14.1. UN number

UN1738

14.2. UN proper shipping name

BENZYL CHLORIDE

14.3. Transport hazard class(es)

6.1

Subsidiary Hazard Class

8

14.4. Packing group

II

IATA

14.1. UN number

UN1738

14.2. UN proper shipping name

BENZYL CHLORIDE

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14.3. Transport hazard class(es) 6.1
Subsidiary Hazard Class 8
14.4. Packing group II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Benzyl chloride	100-44-7	202-853-6	-	-	X	X	KE-05729	X	X
Propylene oxide	75-56-9	200-879-2	-	-	X	X	KE-24565	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Benzyl chloride	100-44-7	X	ACTIVE	X	-	X	X	X
Propylene oxide	75-56-9	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Benzyl chloride	100-44-7	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 28. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	-
Propylene oxide	75-56-9	-	Use restricted. See item 28. (see link for restriction details) Use restricted. See item 29. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - Carcinogenic (Article 57a) SVHC Candidate list - Mutagenic (Article 57b)

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

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REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Benzyl chloride	100-44-7	Not applicable	Not applicable
Propylene oxide	75-56-9	5 tonne	50 tonne

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Benzyl chloride	WGK3	Krebserzeugende Stoffe - Class II : 0.5 mg/m ³ (Massenkonzentration)
Propylene oxide	WGK3	Krebserzeugende Stoffe - Class III : 1 mg/m ³ (Massenkonzentration)

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Benzyl chloride 100-44-7 (>95)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals

H302 - Harmful if swallowed

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H331 - Toxic if inhaled
H315 - Causes skin irritation
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H340 - May cause genetic defects
H350 - May cause cancer
H373 - May cause damage to organs through prolonged or repeated exposure

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Prepared By

Health, Safety and Environmental Department

Creation Date

02-Feb-2010

Revision Date

25-Jan-2024

Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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End of Safety Data Sheet